HANDBOOK · HIGH-ENERGY PHOTON & ELECTRON BEAMS

Dosimetry Handbook

Reference dosimetry is a repeatable recipe: the same conditions, the same measurements and the same corrections, every time. Follow it like a handbook and every beam calibration becomes traceable, comparable and auditable.

$$D_{w,Q} = M_Q \times N_{D,w,Q_o} \times k_{Q,Q_o}$$

$D_{w,Q}$	M_Q	N_{D,w,Q_o}	k_{Q,Q_o}
DOSE TO WATER	CORRECTED READING	CALIBRATION COEFF.	BEAM QUALITY FACTOR

01 Why use dosimetry like a handbook?

One standard procedure

Fixed reference conditions and a fixed order of steps mean any physicist gets the same answer on the same beam.

Traceable to a standard

Every result links back through the chamber's calibration certificate to a primary standards laboratory.

A record you can look up

Each worksheet captures inputs, factors and dose, so past calibrations can be compared, audited and trended.

02 The traceability chain

PRIMARY LAB	SECONDARY LAB	YOUR DOSIMETER	CLINICAL BEAM
PSDL	**SSDL**	N_{D,w,Q_o}	$D_{w,Q}$
Primary standard of absorbed dose to water	Calibrates your chamber in ^{60}Co (Q_o)	Chamber + electrometer with certificate	Dose per MU in your linac beam (Q)

The calibration coefficient is valid only for the ^{60}Co beam it was measured in. The beam quality factor k_{Q,Q_o} carries it over to your accelerator's beam.

How to use this handbook. Page 2 is the procedure: the six steps and every correction factor. Page 3 is the reference: conditions, the sign-off checklist and a glossary of symbols. Each step matches a section of the Interactive Dosimetry Worksheet.

03 The handbook recipe: six steps

1 Set reference conditions
Water phantom, reference field size, SSD or SAD and depth exactly as the code of practice specifies (page 3).

2 Specify the beam quality, Q
Photons: measure $TPR_{20,10}$. Electrons: measure R_{50}, then set $z_{ref} = 0.6\ R_{50} - 0.1$ g/cm².

3 Prepare the dosimeter
Record chamber and electrometer models and serial numbers, N_{D,w,Q_o}, and the lab's T_0 and P_0. Warm up and pre-irradiate.

4 Take the readings
Record T and P, M_1 at your normal voltage V_1, readings at both polarities (M_+, M_-) and M_2 at a lower voltage V_2.

5 Apply the corrections
$M_Q = M_1 \times k_{TP} \times k_{elec} \times k_{pol} \times k_s$, then look up k_{Q,Q_o} for your chamber at beam quality Q.

6 Calculate, compare, record
Compute $D_{w,Q}$ (Gy/MU) and dose rate, compare with the last calibration and expected output, then sign and file the report.

04 Correction factors at a glance

Air density, k_{TP} — T & P

$$k_{TP} = \frac{273.2 + T}{273.2 + T_0} \times \frac{P_0}{P}$$

The chamber is open to air; this normalises the mass of air in the cavity back to calibration conditions.

Polarity, k_{pol} — POLARITY EFFECT

$$k_{pol} = \frac{|M_+| + |M_-|}{2\,M}$$

M is the reading at your usual polarity. Expect a value very close to 1; a larger deviation is worth investigating.

Recombination, k_s — LOST IONS

$$k_s = a_0 + a_1(M_1/M_2) + a_2(M_1/M_2)^2$$

Two-voltage method for pulsed beams. The coefficients a_i come from the tabulated voltage ratio V_1/V_2.

Electrometer, k_{elec} — ELECTROMETER CAL.

$$k_{elec} = \text{from certificate}$$

Equals 1.000 when the chamber and electrometer were calibrated together as one system.

Beam quality, k_{Q,Q_o} — BEAM ≠ CALIBRATION BEAM

$$k_{Q,Q_o} = f(\text{chamber model, } TPR_{20,10} \text{ or } R_{50})$$

Looked up from the code-of-practice tables for your chamber type. It converts the ^{60}Co calibration coefficient to your accelerator's beam quality.

05 Reference conditions quick-look

QUANTITY	HIGH-ENERGY PHOTONS	HIGH-ENERGY ELECTRONS
Beam quality index	$TPR_{20,10}$	R_{50} (g/cm²)
Phantom	Water	Water
Reference depth, z_{ref}	10 g/cm² (or 5 g/cm²)	0.6 R_{50} − 0.1 g/cm²
Distance	SSD or SAD of 100 cm	SSD of 100 cm
Field size	10 cm × 10 cm	≥ 10 × 10 cm² (R_{50} ≤ 7 g/cm²), else ≥ 20 × 20 cm²
Chamber	Cylindrical	Plane-parallel (cylindrical allowed for R_{50} ≥ 4 g/cm²)

06 Checklist before you sign off

- ☐ Calibration certificate is in date and matches the serial numbers.
- ☐ Readings are stable and repeated, not single shots.
- ☐ k_{Q,Q_o} uses the right chamber model and beam quality.
- ☐ Report is printed, signed and filed with the date and user.
- ☐ Thermometer has reached equilibrium with the water.
- ☐ Every correction factor is close to 1; question any outlier.
- ☐ Result agrees with the previous calibration and expected output.
- ☐ Any deviation is investigated before clinical use.

07 Glossary of symbols

$D_{w,Q}$	Absorbed dose to water in beam quality Q	M_Q	Dosimeter reading corrected for all influence quantities
N_{D,w,Q_o}	Calibration coefficient in the reference beam Q_o (^{60}Co)	k_{Q,Q_o}	Beam quality correction from Q_o to Q
$TPR_{20,10}$	Tissue-phantom ratio at 20 and 10 g/cm²; photon beam quality	R_{50}	Depth of 50% absorbed dose; electron beam quality
z_{ref}	Reference depth of measurement in water	T_0, P_0	Temperature and pressure at calibration
M_1, M_2	Readings at the normal voltage V_1 and lower voltage V_2	M_+, M_-	Readings at positive and negative polarity

Educational summary. This handbook is a teaching and quick-reference aid. Always follow the full IAEA TRS-398 text, your chamber's calibration certificate and your local protocols for clinical calibration.